# 52 The Pathophysiology of Uremia

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CHAPTER OUTLINE



The word "uremic" is generally used to describe those ill effects of kidney failure that we cannot yet explain. Hypertension caused by volume overload, tetany caused by hypocalcemia, and anemia caused by erythropoietin deficiency were once considered uremic signs, but were removed from this category as their causes were discovered. Uremia may thus now be defined as the illness that would remain if the extracellular volume and inorganic ion concentrations were kept normal and the known renal synthetic products were replaced in patients without kidneys (Box 52.1).

Some features of uremia, thus defined, could reflect the lack of unidentified renal synthetic products. We presume, however, that uremia is caused largely by the accumulation of organic waste products that are normally cleared by the kidneys. In general, the study of renal organic waste removal lags far behind the study of inorganic ion excretion. A major problem is the multiplicity of waste solutes. The first comprehensive review prepared by the European Uremic Toxin Work Group (EUTox[1,2]) listed >80 uremic solutes, and further studies have increased this number to >250.[3–7] Untargeted mass spectrometry has revealed that the true number is much greater, and that the majority of uremic solutes cannot be found in standard databases of known biological compounds. With so many substances to study, it is hard to establish which ones are toxic. Bergstrom[8] suggested criteria for identifying uremic toxins that are analogous to Koch postulates for identifying infectious agents. According to these criteria, a uremic toxin must have a known chemical structure and the following features:

- Its plasma and/or tissue concentrations should be higher in patients with kidney failure than in normal persons.
- The high concentrations should be related to specific uremic symptoms that are ameliorated when the concentration is reduced.
- The effects observed in uremic patients should be replicated by raising the solute concentration to similar levels in normal persons, experimental animals, or in vitro systems.

No uremic solute has so far been shown to fully satisfy these criteria. Given the complexity of uremia, it is unlikely that the accumulation of a single solute in isolation could recapitulate uremia or that removal of the same could eliminate uremia; therefore progress in uremia research has been relatively slow. Most solutes that accumulate with advanced chronic kidney disease (CKD) are probably not toxic. Others that are toxic may exert their ill effects only when administered in combination. The difficulty imposed by the multiplicity of solutes is compounded by the multiplicity of ill effects encountered in uremia. Investigators of uremic toxicity thus face the daunting task of matching a solute or group of solutes to an appropriate endpoint. Many of the effects of uremia are hard to quantify, which makes the problem even more difficult. This is particularly true of major uremic symptoms, especially fatigue, anorexia, and diminished mental acuity.

A related problem encountered in studies of uremia is distinguishing the effects of uremia from those of related conditions. Paradoxically, the widespread availability of dialysis has made uremia more difficult to study. The severity of the classic uremic symptoms is attenuated, and patients now suffer from a new illness, which Depner[9] has named the "residual syndrome," which is composed of partially treated uremia and the side effects of dialysis. In most patients, features of the residual syndrome are further combined with the effects of age and of systemic diseases responsible for kidney failure. Disturbances of inorganic ion metabolism, including acidemia and hyperphosphatemia, although excluded from our definition of uremia, undoubtedly also contribute to untoward clinical manifestations of kidney failure. Given these difficulties, it is not surprising that although we have identified many uremic solutes, we know relatively little about their toxicity. In a few cases, uremic abnormalities

**Box 52.1 Metabolic Effects, Symptoms, and Signs of Uremia**

**Metabolic**

Increased oxidant levels
Reduced resting energy expenditure
Reduced body temperature
Insulin resistance
Muscle wasting
Amenorrhea and sexual dysfunction

**Neural and Muscular**

Fatigue
Loss of concentration ranging to coma and seizures
Sleep disturbances
Restless legs
Peripheral neuropathy
Anorexia and nausea
Diminution in taste and smell
Itching
Cramps
Reduced muscle membrane potential

**Other**

Serositis (including pericarditis)
Hiccups
Granulocyte and lymphocyte dysfunction
Platelet dysfunction
Shortened erythrocyte life span
Albumin oxidation

**Box 52.2 Uremic Abnormalities Transferable With Uremic Serum or Plasma**

Inhibition of $Na^{+}$,$K^{+}$-ATPase
Inhibition of platelet function
Leukocyte dysfunction
Loss of erythrocyte membrane lipid asymmetry
Insulin resistance

*ATPase,* Adenosine triphosphatase.

have been reproduced by the transfer of uremic serum or plasma to normal animals or addition of these factors to the media of cultured cells (Box 52.2). However, the role of particular solute(s) in causing the abnormalities remains uncertain.

## SOLUTES CLEARED BY THE KIDNEY AND RETAINED IN UREMIA

The long list of solutes retained in uremia has been assembled in two ways. Initially, biochemists would find a substance in the urine and then look for it in the blood of uremic patients. Several dozen uremic solutes were identified in this way as the biochemical pathways of intermediary metabolism were worked out. Beginning about 1970, improved analytic techniques, including gas chromatography, mass spectroscopy, and high-performance liquid chromatography, were used to identify additional uremic solutes.[3,10] Recent technical advances, including proteomic and metabolomic screening methods, are lengthening the list of putative uremic solutes. However, the problem is to determine which solutes are toxic. In general, the compounds that are present in the highest concentrations, and therefore had been identified first, have been studied most extensively. Experiments showing that uremic signs and symptoms can be replicated by raising solute levels in normal persons or animals to equal those observed in uremic patients are, for the most part, lacking. When attempted, such experiments have generally demonstrated that the solutes being studied are more toxic than urea, but that the levels required to produce toxic effects are higher than those measured in patients. Because so little is known about their toxicity, the discussion of uremic solutes is usually organized on the basis of their structure and not their contribution to disease.

## INDIVIDUAL UREMIC SOLUTES

### UREA

Urea is quantitatively the most abundant solute excreted by the kidney, and its levels rise higher than those of any other solute when the kidney fails. Early studies indicated that urea causes only a minor part of uremic illness.[11–13] In the most often cited of these studies, Johnson and colleagues[12] dialyzed patients with kidney failure against bath solutions containing urea. They found that initiation of hemodialysis improved uremic symptoms, including weakness, fetor, and gastrointestinal upset, even when urea was added to the dialysate to maintain the blood urea nitrogen (BUN) level at approximately 90 mg/dL (32 mmol/L). In patients already on dialysis, increasing the BUN level to 140 mg/dL (50 mmol/L) did not cause recurrence of uremic symptoms. Increasing the BUN level >140 mg/dL caused nausea and headaches, and increasing the BUN level >180 mg/dL (64 mmol/L) caused weakness and lethargy. However, symptoms in dialyzed patients whose BUN values were increased to these levels were less severe than symptoms in undialyzed patients with high BUN values. Studies in patients without kidney failure suggest that urea by itself does not cause uremia. Uremic symptoms have not been observed in patients in whom BUN levels are maintained at approximately 60 mg/dL (21 mmol/L) by high protein intake or increased tubular urea absorption.[14–16] Similarly, patients on high-dose glucocorticoids or those with heart failure, conditions commonly seen in modern medical practice, do not experience uremia when kidney function, estimated by other solutes (typically serum creatinine), is not severely impaired.

The finding that uremia is not replicated by an isolated elevation of the plasma urea concentration does not mean that urea has no toxic effects.[17] The full expression of uremia may require accumulation of urea plus other solutes. Johnson and associates[12] noted that patients dialyzed against solutions of urea exhibited increased bleeding, and subsequent studies suggested that urea promotes bleeding by promoting synthesis of guanidinosuccinic acid (GSA), which impairs platelet

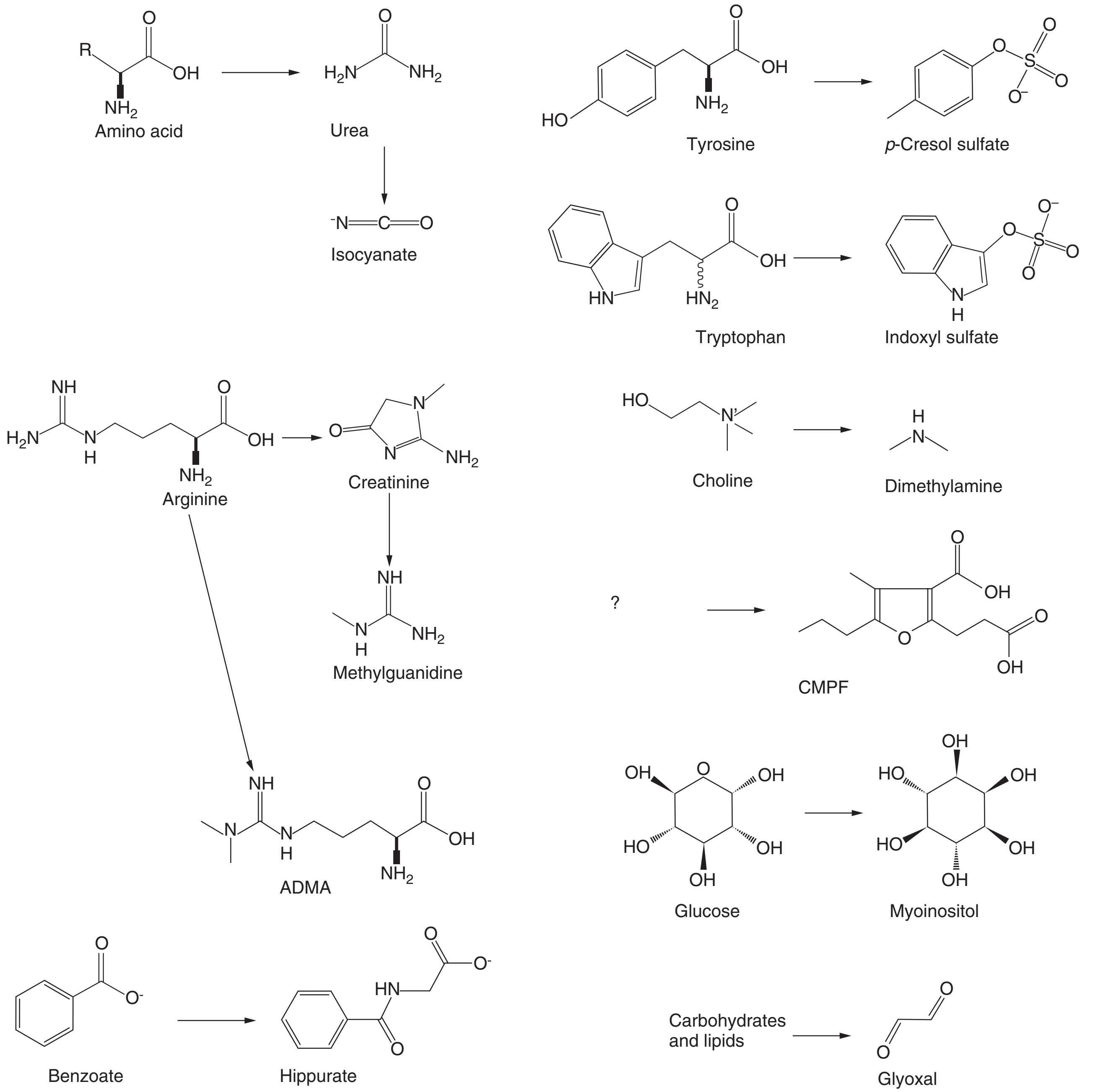


**Fig. 52.1** Generation of potential uremic toxins. The substances in the *right column* of each panel are metabolites that are normally excreted by the kidney and therefore accumulate in the extracellular fluid when kidney function is lost. The *left column* shows the substances from which these potential "uremic toxins" are derived. In some cases, the biochemical derivation of the potential toxins is uncertain. For example, it is not known what fraction of the dimethylamine normally excreted is derived from choline, and the source of 3-carboxy-4-methyl-5-prophy-2-furanpropanoic acid (CMPF) is obscure. See text for details. *ADMA,* Asymmetric dimethyl arginine.

function.[18,19] Increased plasma urea concentrations may cause other ill effects by increasing isocyanate concentration and thereby promoting protein carbamylation (Fig. 52.1).[20–24] Isocyanate can also combine reversibly with –OH and –SH groups of amino acids, and the various isocyanate-induced alterations in structure could impair protein function. A prominent contribution of urea to uremic toxicity would, however, be difficult to reconcile with the lack of benefit detected when urea levels were reduced by more intensive dialysis in the Hemodialysis (HEMO) study.[25]

## D–AMINO ACIDS

In comparison with urea, we know much less about most other potential uremic toxins. The D–amino acids exemplify this problem. Aggregate plasma concentrations of D–amino acids increase as kidney function declines.[26,27] However, the source, clearance, and toxicity of the D–amino acids found in the plasma are not well defined. D–Amino acids can be synthesized by mammalian cells, as well as derived from food, or produced by colonic bacteria.[28] Circulating D–amino acids are filtered by the glomerulus and then, in varying proportion, reabsorbed intact, degraded by D–amino acid oxidase (DAO) or D–aspartic acid oxidase in the proximal straight tubule, or excreted unaltered in the urine.[29,30] The liver can also clear D–amino acids, but the relative importance of renal and hepatic clearance is unknown. Aggregate D–amino acid concentrations have been found to increase almost in proportion to the serum creatinine level in kidney failure, suggesting that renal clearance predominates.[26,31,32] However, measured concentrations of individual D–amino acids, such as D-serine, increase less than creatinine.[31,32] This discrepancy remains unexplained. Of interest, D-serine is an endogenous coagonist of the synaptic *N*-methyl-D-aspartate receptor. It is tempting to speculate that D–amino acids are cleared rapidly from the extracellular fluid (ECF) because they have toxic effects. In addition, it has long been presumed that high levels of D–amino acids could impair protein synthesis or function.[28] D–amino acid accumulation could also interfere with the effects of endogenous D-serine and D-alanine on neuronal function,[33] but no major ill effects of D–amino acid accumulation have been observed in DAO-deficient mice, which have higher D–amino acid levels than humans with impaired kidney function.[34,35] Exogenous D–amino acids have so far been shown to be toxic only when administered in large quantities.[34,36]

## PEPTIDES AND PROTEINS

The kidney clears circulating dipeptides and tripeptides, which may comprise a significant portion of the extracellular amino acid pool.[37] Filtered dipeptides and tripeptides can be broken down by brush-border peptidases and reabsorbed as amino acids or reabsorbed by a brush-border peptide transporter and then hydrolyzed within proximal tubule cells.[38] Peritubular uptake, again followed by hydrolysis to amino acids, makes the renal clearance of many peptides higher than the glomerular filtration rate (GFR).[37,39] Small peptides are also taken up by other organs and generally do not accumulate in kidney failure. Peptides containing altered amino acids, which are normally cleared by the kidney, may be an exception to this rule.[39]

The kidney plays a proportionally larger role in the clearance of larger peptides. Proteins with a molecular weight of 10 to 20 kDa, such as $\beta_2$-microglobulin and cystatin C, are normally filtered by the glomerulus and then endocytosed and hydrolyzed in the lysosomes of proximal tubular cells.[40,41] Their plasma concentrations therefore rise in parallel with the plasma creatinine level as kidney function declines. Indeed, the plasma concentration of cystatin C, which is released at a near-constant rate by nucleated cells, may yield a more reliable estimate of GFR than the concentration of creatinine.[42] The role of the kidney in the removal of peptides with molecular weight between 500 Da and 10 kDa is less well defined. Peptides in this range are also filtered by the glomerulus and then hydrolyzed by brush-border peptidases or endocytosed, depending on their size and structure. Biologically active peptides such as insulin may also be cleared by peritubular uptake. Studies in patients with inherited dysfunction of proximal tubular endocytosis suggest that the normal kidney clears approximately 350 mg/day of peptides with molecular weights of 5 to 10 kDa from the circulation.[43] The relative importance of renal to extrarenal clearance has not been defined for most substances in this size range. The extent to which circulating levels of such peptides are increased in kidney failure is therefore unpredictable. Even less is known about the kidneys' contribution to the clearance of peptides in the range of 500 Da to 5 kDa.

**Box 52.3 Low-Molecular-Weight Proteins and Protein Fragments That Accumulate in Uremia**

$\alpha_1$-Microglobulin
$\beta_2$-Microglobulin
$\beta$-Trace protein (also known as prostaglandin D2 synthase)
Clara cell protein
Chromogranin A
Cystatin C
Free immunoglobulin light chains
Natriuretic peptides
Procalcitonin
Retinol-binding protein
Transcobalamin
Leptin
Ghrelin
Resistin
Troponin I

Although the aggregate peptide levels in kidney failure remain ill defined, we have some knowledge of individual peptides that are retained in uremia. The best known are small proteins or fragments of large proteins for which immunoassays have been developed.[44–58] Box 52.3 provides a partial list of these substances. Retinol-binding protein, $\alpha_1$-microglobulin, and $\beta$-trace protein (also known as prostaglandin $D_2$ synthase) are members of the lipocalin superfamily, and future studies may identify elevated levels of other proteins of this group. Lipocalins are a family of proteins that transport small hydrophobic molecules and have an eight-stranded, antiparallel, symmetrical $\beta$-barrel fold. This reflects a $\beta$ sheet that has been rolled into a cylindrical shape. A ligand-binding site is present in this cylinder. Studies using proteomic techniques have yielded a more complete picture of individual peptides that are retained in uremia.[59–61] These studies have shown that, as expected, uremic plasma contains a vast array of protein fragments that are normally cleared by the kidney. Many of these are derived from fibrinogen and the complement cascade.[61,62] One study has identified >1000 peptides with molecular weights from 800 Da to 10 kDa in the plasma of patients undergoing dialysis.[63] The toxicity of these peptides and low-molecular-weight proteins, like that of smaller molecular uremic solutes, is largely unknown. It has been widely speculated that retained peptides can cause inappropriate activation of various hormone or cytokine receptors. For example, retained complement protein D (molecular weight, 24 kDa) could contribute to the systemic inflammation and accelerated vascular disease observed in patients receiving dialysis.[64] Such hypotheses remain, however, to be proven. $\beta_2$-Microglobulin amyloidosis is the only disease attributable with certainty to a retained peptide, and its prevalence among the dialysis population appears to have declined over the past 30 years.[65] Plasma levels of peptides such as procalcitonin, troponin I, N-terminal brain natriuretic peptide, and chromogranin A are, however, increasingly used as diagnostic markers. And "false-positive" results may be obtained if criteria are not adjusted for the elevation of their plasma levels which normally accompanies loss of renal function.

## GUANIDINES

Among the compounds most frequently considered uremic toxins are guanidines, which, like urea, are derived from arginine (see Fig. 52.1).[66–68] One group of guanidines that accumulate in uremia includes creatinine and its breakdown products. Creatinine is produced by nonenzymatic degradation of creatine, which in turn is made from guanidinoacetic acid.[69] Creatinine itself appears nontoxic, and levels have been increased transiently to >100 mg/dL (8800 µmol/L) in subjects undergoing clearance studies. Instead, interest has been focused on the potential toxicity of various creatinine metabolites, especially creatol and methylguanidine (MG).[70,71] Creatol is a product resulting from the reaction of creatinine with the hydroxyl radical and is identified as a precursor of MG. Creatol and MG levels normally approximate 1% of creatinine levels. The production of these substances increases as plasma creatinine concentrations rise and may be stimulated by increased levels of intracellular oxidants.[67,69,70] MG is also produced by colonic bacteria, and its production may be increased by augmented dietary intake of protein or creatinine.[72] Another guanidine that has attracted interest is GSA, a substance formed not from creatinine but from the urea cycle intermediate argininosuccinate.[73,74] Rising plasma urea concentrations impede the conversion of argininosuccinate to urea and increase the production of GSA. The production of GSA thus depends on dietary protein intake as well as on kidney function, and may also be stimulated by increased concentrations of intracellular oxidants.[74,75]

Creatol, MG, and GSA share the interesting property that their plasma concentrations rise out of proportion to urea and creatinine levels as the GFR declines. This is because they are cleared largely by the kidney, and their production rates increase when plasma creatinine and urea concentrations are elevated.[67,69,70] In addition, relative to creatinine, large volumes of distribution, combined with restricted intercompartmental diffusion, limit the removal of creatol, MG, and GSA by intermittent hemodialysis.[66] In patients undergoing conventional hemodialysis, these compounds therefore exhibit high concentrations relative to normal.[1] The finding that they are present in relatively high concentrations does not prove that they are toxic. However, the evidence for toxicity of various guanidines, although incomplete, is stronger than that for most other solutes. Administration of MG aggravates uremic symptoms in dogs, whereas GSA contributes to uremic platelet dysfunction, and a number of guanidines impair neutrophil function.[18,76,77] In addition, various guanidines have been shown to accumulate in the brain and cerebrospinal fluid in uremia and may contribute to central nervous system dysfunction.[78]

The methylated arginines, asymmetric dimethyl arginine (ADMA) and symmetric dimethyl arginine (SDMA), also accumulate in kidney failure (Fig. 52.1). The metabolism of methylated arginines is quite different from that of the other uremic guanidines. ADMA and SDMA are formed by the methylation of arginine residues in nuclear proteins and released when these proteins are degraded. Interest has focused largely on ADMA because it inhibits nitric oxide (a potent local vasodilator) synthesis by the endothelial nitric oxide synthase (eNOS). The enzyme eNOS converts L-arginine to nitric oxide and L-citrulline and is an endothelial-enriched messenger RNA and protein. ADMA inhibits eNOS enzymatic activity.[79,80] SDMA has been less extensively considered as a toxin, but recent evidence has associated its plasma levels with cardiovascular risk.[80] The urinary clearance of ADMA is similar to that of creatinine, but most plasma ADMA is taken up and degraded in various tissues, including the kidney.[79,81] The presence of nonrenal clearance limits the rise in plasma ADMA concentrations observed as kidney function declines, so that ADMA concentrations rise to approximately two times normal early in the course of CKD and then do not increase much further as patients advance to end-stage renal disease (ESRD).[81,82] Increases in plasma ADMA concentration, although modest in proportion to other uremic solutes, have been associated with an increased risk for cardiovascular disease and death in patients with CKD.[80] It should be noted that differences in assay methods and reported reference ranges for ADMA greatly complicate the interpretation of these studies.

## PHENOLS AND OTHER AROMATIC COMPOUNDS

Phenols are compounds that have one or more hydroxyl groups attached to a benzene ring. In discussions of uremia, phenols are usually considered together with other aromatic compounds, such as hippurates, and the term “phenols” is sometimes used loosely to include these other substances. The aromatic compounds normally found in ECF are, for the most part, derived from the amino acids tyrosine and phenylalanine or from aromatic compounds contained in dietary vegetables. Medications provide an additional source in patients. The compounds in ECF are mostly metabolites; these are derived from their parent compounds by a combination of methylation, dehydroxylation, oxidation, reduction, and/or conjugation. Many of these reactions take place in colonic bacteria. The final step, which is usually conjugation with sulfate, glucuronic acid, or an amino acid, may take place in the liver, intestinal wall, or, to a lesser extent, kidney.[83,84] In general, conjugation tends to make the aromatic compounds both less toxic and more polar, thus facilitating their excretion by various organic ion transport systems.

These metabolic processes produce a bewildering array of aromatic compounds that are normally excreted in the urine or feces. The aggregate urinary excretion of aromatics is about 1000 mg/day and varies widely with the diet. The compounds normally excreted by the kidney accumulate in uremia and contribute to elevation of the anion gap, because most aromatic conjugates are negatively charged.[85] The concentration of individual aromatic compounds in uremic patients ranges from barely detectable up to 500 µM.[1,86–88] The relatively few compounds that have been studied extensively, including the examples described later, are among those found in the highest concentration. Interest in the contribution of phenols and other aromatic compounds to uremic toxicity has been encouraged by reports that uremic symptoms are better correlated with plasma concentration of these compounds than with those of other solutes.[13,89–91] Evidence obtained so far on the toxicity of individual aromatic compound is incomplete.

The most extensively studied aromatic uremic solute is hippurate (see Fig. 52.1). Because it is the aromatic waste compound normally excreted in the largest quantity, the concentration of free hippurate rises higher than those of other aromatic solutes in the plasma of patients with kidney failure. Hippurate is the glycine conjugate of benzoate, derived largely from vegetable foods, with only a small amount formed endogenously from the amino acid phenylalanine.[92,93] Diet therefore determines hippurate production, and hippurate excretion in aboriginal people eating vegetable diets may exceed hippurate excretion in people from industrialized nations by many fold.[94] In persons with normal kidneys, active tubular secretion maintains a plasma hippurate concentration much lower than it would be if hippurate were cleared solely by glomerular filtration. There is a clinical correlate of this property of hippurate. Aminohippuric acid or *para*-aminohippuric acid (PAH), a derivative of hippuric acid, is a diagnostic agent used in the measurement of renal plasma flow. PAH is completely removed from the blood that passes through the kidneys given that PAH undergoes both glomerular filtration and tubular secretion. Therefore the rate at which the kidneys can clear PAH from the blood reflects total renal plasma flow. By itself, hippurate is not toxic. The plasma hippurate concentration in a normal person can be increased to equal that of a uremic patient, without apparent ill effect.[95] Moreover, increasing the hippurate concentration by benzoate feeding in a patient with kidney failure does not aggravate uremic symptoms.[96] A further clinical correlate addresses the potential toxicity of aromatics, in this case an exogenous source. Toluene is a widely abused inhaled drug (e.g., glue sniffing) due, in part, to its acute neurologic effects of euphoria and hallucinations. Toluene is metabolized by cytochrome P-450 into benzoic acid and hippuric acid. The hallmarks of acute toluene intoxication are hypokalemic paralysis and metabolic acidosis due to distal renal tubular acidosis. Patients can also have hippurate crystals appear in the urine.

Another extensively studied aromatic compound is *p*-cresol. In contrast to hippurate, which is derived from aromatic compounds in plants, *p*-cresol is formed by the action of colonic bacteria on tyrosine and phenylalanine. The portion of amino acids that escapes absorption in the small intestine may be increased in uremic patients, leading to increased production of *p*-cresol and other bacterial metabolites.[97,98] Studies have shown that *p*-cresol circulates largely as *p*-cresol sulfate with a much lesser portion in the form of *p*-cresol glucuronide. Reports of unconjugated *p*-cresol in the plasma of uremic patients now appear to have been the result of inadvertent hydrolysis of *p*-cresol sulfate during sample processing.[99,100] *p*-Cresol sulfate binds avidly to serum albumin, and the effect of different renal replacement therapies on albumin-bound solutes has often been tested by measuring plasma *p*-cresol sulfate concentrations.[101–103] High concentrations of *p*-cresol sulfate (often measured as *p*-cresol) have been associated with cardiovascular death in patients undergoing hemodialysis, and *p*-cresol sulfate has been related to indices of endothelial injury, including endothelial microparticle production.[104–106] While the results of clinical studies are not uniform, these findings, along with in vitro evidence of *p*-cresol sulfate toxicity, have increased the focus on *p*-cresol sulfate as a potential uremic toxin.[105–107]

Other aromatic uremic solutes have been identified in great numbers but studied less extensively.[8,87,88] Metabolites of tyrosine and phenylalanine, which accumulate in uremia, include phenlylacetylglutamine, *para*-hydroxyphenylacetic acid, and 3,4-dihydroxybenzoic acid, as well as *p*-cresol.[108–110] Phenylacetylglutamine, like *p*-cresol sulfate, has been associated with cardiovascular disease and mortality in dialysis patients.[111] The structural relationship of the aromatic amino acid metabolites to neurotransmitters has stimulated interest in their potential role as uremic toxins. So far, 3,4-dihydroxybenzoate has been shown to cause central nervous system dysfunction in rats, but only at levels higher than those encountered in patients with kidney failure.[109] Increased levels of 4-hydroxyphenylacetate were associated with impaired cognitive function in patients maintained on hemodialysis.[112] The work of identifying toxic aromatic uremic solutes, however, remains daunting. Relatively little progress has been made, and caution is appropriate, particularly when interpreting studies relating levels of specific compounds to poor outcomes. Associations have necessarily been identified only for the few compounds extensively studied. Even if a positive association has been correctly identified, correlations of poor outcomes with abnormal blood levels could be confounded by associations with other unmeasured aromatic substance(s) evidencing related production and/or clearance rates.

## INDOLES AND OTHER TRYPTOPHAN METABOLITES

Indoles are compounds containing a benzene ring fused to a five-membered nitrogen-containing pyrrole ring (Fig. 52.1). Many similarities are encountered when considering the indoles and phenols in uremia. As with phenols, some indoles are derived from plant foods, and others are produced endogenously. However, the endogenous indoles are derived mostly from tryptophan, whereas the phenols are derived from phenylalanine and tyrosine. As with the phenols, minor chemical modifications in various combinations yield a remarkable variety of structures, with more than 600 indoles derived from tryptophan.[113] Those with known physiologic function include the neurotransmitter 5-hydroxytryptamine (serotonin) and melatonin. Other indoles are considered to be waste products and are often conjugated prior to urinary excretion. These uremic indoles accumulate when kidney function is impaired.

The most extensively studied of the uremic indoles is indoxyl sulfate; this is produced from tryptophan in a manner reminiscent of the production of *p*-cresol sulfate from tyrosine and phenylalanine. Gut bacteria convert tryptophan to indole, which is then oxidized to indoxyl and conjugated with sulfate in the liver. There is evidence that indoxyl sulfate is toxic in vitro, but early studies of indoxyl sulfate infusion failed to replicate uremic symptoms.[13,114–116] Like *p*-cresol sulfate, indoxyl sulfate is extensively bound to plasma albumin. It has also been suggested that indoxyl sulfate is toxic to renal tubular cells and the rising indoxyl sulfate levels accelerate the progression of renal disease.[117] Controlled trials to lower indoxyl sulfate levels, however, have not been found to slow progression.[118]

Other indoles that accumulate in uremia include indoleacetic acid, indoleacrylic acid, and 5-hydroxyindoleacetic acid.[1,119,120] As with the phenols, indoles are structurally related to potent neuroactive substances, including serotonin and (famously) D-lysergic acid diethylamide. This structural similarity has stimulated interest in the potential role of indoles as neurotoxins, but few uremic indoles have been administered to normal animals, and none have convincingly been shown to alter CNS function at the levels encountered in patients with kidney failure. There is, however, increasing evidence that indoxyl sulfate contributes to vascular disease and particularly to thrombosis.[106,121] Some effects of indoxyl sulfate have been shown to be mediated by activation of the aryl hydrocarbon receptor, a transcription factor originally identified as upregulating xenobiotic disposal pathways.[121,122] This work puts our knowledge of the mechanisms by which indoxyl sulfate acts ahead of that for other putative uremic toxins.

Only a minor portion of dietary tryptophan is excreted as indoles. Most is metabolized by the kynurenine pathway, which allows tryptophan to be converted to glutarate and oxidized or, when necessary, used in the synthesis of nicotinamide. Kidney failure causes members of the kynurenine pathway, including L-kynurenine and quinolinic acid, to accumulate in the plasma.[123,124] Knowledge that these substances play a physiologic role in the modulation of CNS function has stimulated interest in their possible contribution to uremic toxicity. As usual, however, evidence that they are toxic at the levels encountered in patients has not been obtained.

## ALIPHATIC AMINES

The methylamines monomethylamine (MMA), dimethylamine (DMA), and trimethylamine (TMA) are among the simplest compounds that have been considered to be uremic toxins. Reported serum levels are twofold to threefold higher in patients with ESRD compared with persons with normal or near-normal kidney function.[125,126] However, available data and predictions based on their chemistry suggest that the methylamines are poorly removed by dialysis, and early data suggest that they may even be produced in excess in those with uremia.[125–127]

A large volume of distribution may contribute to poor removal of the methylamines by dialysis. These compounds are bases, with a $pK_a$ ranging from 9 to 11. Thus they exist as positively charged species at a physiologic pH. The lower intracellular pH compared with extracellular pH should lead to their preferential intracellular sequestration, with volumes of distribution exceeding total body water. Indeed, measurements in experimental animals and humans have confirmed these predictions for DMA and TMA.[126,128,129]

Because they circulate as small organic compounds that are not protein bound, these three amines are likely freely filtered. However, because they exist as organic cations, they also have the potential to be secreted by one or another of the family of organic cation transporters and may also travel through Rh channels.[130,131] Hence they may achieve clearances that are higher than the GFR. The chemically similar exogenous compound, tetraethyl ammonium, has long been a prototype test solute for organic cation secretion and is cleared at rates up to (and in one study higher than) the renal plasma flow.[132,133] Although formal renal clearances of DMA and TMA are not available, the total metabolic clearance of DMA and TMA by plasma disappearance of labeled compounds in rats approaches that of renal plasma flow.[128] By contrast, the urinary clearance of MMA in normal subjects is about one-third that of creatinine, indicating no net secretion for this amine.[127]

The biochemical pathways leading to MMA, DMA, and TMA are not well delineated. Both the host's mammalian tissues and resident gut flora have been thought to contribute to the net appearance of these amines. However, plasma MMA and DMA concentrations were not different among patients with ESRD with and without colons.[134] The dietary precursors for MMA, DMA, and TMA include choline and trimethylamine oxide (TMAO).[135–137] Production of these compounds may actually be increased with kidney failure, perhaps caused by overgrowth of intestinal bacteria.[127,129,138] Thus production of aliphatic amines may be increased in the face of impaired renal removal.

Incomplete data also implicate the amines as toxic. Reported effects of MMA include a variety of neural toxicities, hemolysis, and inhibition of lysosomal function.[139] MMA has also been shown to be a potent anorectic agent when administered into the cerebrospinal fluid in mice.[140] Despite toxicities in cells and animals, however, MMA and DMA were not associated with all-cause mortality in a cohort of patients with ESRD.[141] Although of utmost importance, mortality is obviously a blunt metric of uremic toxicity. Other signs of toxicity of MMA and DMA should be explored.

The uremic fetor or fishy breath noted in uremic patients is attributable to TMA.[142] Although the malodor may be of no major consequence in itself, the potentially important and well-described diminutions in taste (dysgeusia) and smell (dysosmia) among patients with kidney failure (which may contribute to poor nutritional status) may also be related to the amines. Plasma MMA concentrations were not related to olfactory defects in ESRD.[143]

TMA can be both a precursor to TMAO and, as noted, a product of dietary TMAO.[144] TMAO is cleared by secretion in the normal kidney; its levels rise as the GFR falls and, like those of many other secreted solutes, are increased out of proportion to urea levels in dialysis patients.[145] TMAO was initially identified as a risk factor for cardiovascular disease in persons with normal renal function.[144,146] TMAO is a gut microbial metabolite that has been shown to be directly atherogenic. TMAO arises from gut microbiota metabolism following ingestion of diets rich in phosphatidylcholine (lecithin), the major dietary source of choline, and carnitine, an abundant nutrient in red meat. Recent studies employing microbial transplantation into recipients confirmed a direct causal role for gut microbes in transmitting atherosclerosis susceptibility and overall TMAO production. Subsequent studies showed that accumulation of TMAO is also associated with atherosclerosis burden and cardiovascular events in patients with renal insufficiency and in patients maintained on dialysis.[147,148]

## OTHER UREMIC SOLUTES

A wide variety of other compounds accumulate in kidney failure. One group is the polyols, of which the most extensively studied is myoinositol (Fig. 52.1).[149,150] Myoinositol is different from most other uremic solutes in that it is normally oxidized by the kidney. Its accumulation in uremia therefore reflects impaired degradation and not impaired excretion. Evidence that myoinositol causes nerve damage, although stronger than most of the evidence for the toxicity of uremic solutes, is far from conclusive.[151]

The purine metabolite uric acid is the only known organic substance with a plasma level actively regulated by variation of its renal excretion. With advanced CKD, the capacity of the kidney to increase the fractional excretion of uric acid is exceeded, and uric acid levels increase, along with those of its precursor molecules, xanthine and hypoxanthine.

Observational studies have shown that high uric acid levels are associated with increased survival in dialysis patients, perhaps because the plasma uric acid is a marker of nutritional status.[152] Other nucleic acid metabolites excreted by the kidney are produced in much lesser quantities. Many are derived from the modified nucleosides contained in transfer RNAs.[153] They appear to be cleared largely by filtration and to accumulate in the plasma as the GFR falls. Pseudouridine is an isomer of the nucleoside uridine in which the uracil is attached via a carbon–carbon instead of a nitrogen–carbon glycosidic bond. It has been suggested that pseudouridine, which is the most abundant of these variant nucleic acid metabolites, contributes to insulin resistance and altered CNS development but, as usual, the demonstration of its toxicity is not conclusive.[153,154]

Oxalate is also excreted by the kidney and accumulates in kidney failure. Oxalate is derived from catabolism of endogenous substances, including vitamin C, as well as from plant foods.[155,156] The potential for oxalate deposition in tissues may limit our ability to maintain normal plasma vitamin C concentrations in patients receiving dialysis.[157,158] Additional substances excreted by the kidney that accumulate in kidney failure include various pteridines, dicarboxylic acids, isoflavins, and furancarboxylic acids, including 3-carboxy-4-methyl-5-prophy-2-furanpropanoic acid.[102,153,159–162] Recently completed studies continue to add new solutes to the list, and the number reported will likely soon rise >1000, including many compounds that are at present identified only by their molecular mass and do not appear in standard databases of human metabolites as yet. The possibility of toxicity is invariably considered when new solutes are identified, but experiments to test the solute toxicity are rarely performed.

## SOLUTE REMOVAL BY DIFFERENT FORMS OF RENAL REPLACEMENT THERAPY

Although investigators have not succeeded in replicating uremic illness by administering uremic solutes to normal humans or animals, reversing illness by removing solutes has become a part of everyday practice. Because renal replacement therapies remove solutes indiscriminately, the improvement they effect cannot be attributed to removal of specific compounds. However, different forms of renal replacement therapy do clear solutes at different rates based on some characteristics, including molecular size, protein binding, and sequestration within cells or other body compartments. The demonstration that different therapies have different effects on some features of uremia might therefore reveal properties of the responsible toxin(s).

### ORIGINAL MIDDLE MOLECULE HYPOTHESIS

The suggestion that the nature of uremic toxins could be deduced by comparing the effects of different renal replacement methods was first advanced by Babb and colleagues.[163] In the 1960s, hemodialysis was performed with membranes that provided very limited clearance of solutes with molecular weight >1000 Da. Treatment with these membranes wakened patients from coma, relieved vomiting, and partially reversed other uremic symptoms. This provided evidence, which remains convincing, that some important uremic toxins are small. Babb and coworkers were impressed that patients on peritoneal dialysis were healthier than patients on hemodialysis who had the same plasma urea and creatinine concentrations. They further observed that increasing the dialysis duration from 6.5 to 9 hours three times weekly prevented neuropathy. These observations led them to conclude that important toxins were >300 Da because, as compared with contemporary hemodialysis membranes, the peritoneal membrane afforded greater relative permeability in this size range and because increasing the hemodialysis session length was expected to reduce the plasma concentration of larger molecules more than the concentrations of creatinine and urea. Based on their further impression that no additional benefit was obtained using membranes that provided superior clearance for solutes >2000 Da, they concluded that some important toxins were "middle molecules," with a molecular weight >300 Da but <2000 Da.[164]

### LARGE SOLUTES—CHANGING DEFINITION OF "MIDDLE MOLECULES"

Studies during the 1970s provided only equivocal evidence that increasing the clearance of solutes with a molecular weight between 350 and 2000 Da improved the health of uremic patients.[163] The proposition that no benefit could be obtained by increasing the clearance of solutes with molecular weight >2000 Da was never prospectively tested. The original middle molecule hypothesis was thus never proven to be correct. And although the phrase "middle molecules" remains in use, its meaning has gradually shifted to include larger solutes. The 2003 report of the EUTox work group[1] thus defined middle molecules as those with a size ranging from 500 Da to <60,000 Da. The adoption of new membrane materials, which was in part a response to the original middle molecule hypothesis, ended the investigation of the relative toxicity of solutes that fall into different parts of the size range <1000 Da. The question of whether solutes with molecular weight >1000 Da exert toxic effects remains under investigation. Henderson and colleagues[165] showed that such solutes can be cleared more effectively by hemofiltration than by hemodialysis. The results of recent large trials combining hemofiltration with hemodialysis to increase the clearance of large solutes have been equivocal. One problem is that the concentration of putative high-molecular-weight toxins may not decline in proportion to the increase in clearance during treatment because they are cleared by extrarenal mechanisms and they move slowly from the interstitial fluid to the ECF during treatment.[166,167]

### PROTEIN-BOUND SOLUTES

Other solutes that are poorly removed by standard hemodialysis include those that bind to albumin.[102,168] Their dialytic clearance is low, not because they are large molecules, but because only the free, unbound solute concentration contributes to the gradient-driving solute across the dialysis membrane. In the normal kidney, the combination of protein binding and tubular secretion allows molecules to be excreted while keeping their concentrations in the ECF very low.[5] This presumably represents an evolutionary adaptation to excrete toxic substances, and there is indeed suggestive evidence that some important uremic toxins are protein bound.[169] The clearance of protein-bound solutes can be increased by raising dialysate flow and membrane size above the levels used in conventional hemodialysis or by combining a high hemofiltration rate with dialysis in hemodiafiltration treatment.[170,171] The aggregate toxicity of protein-bound solutes could thus theoretically be assessed by comparing the effects of different renal replacement prescriptions, but this has not been attempted do date. Other methods of improving bound solute removal including the use of sorbents, infusion of compounds that displace bound solute from albumin, and manipulation of the ionic strength of the dialysate remain to be tested. Studies carried out raise the possibility that levels of protein-bound solutes, like levels of large solutes, may not decline in proportion to the increase in clearance during treatment.[167] Peritoneal dialysis clears protein-bound solutes at a very low rate, and the total clearance of protein-bound solutes in patients maintained on peritoneal dialysis therefore depends heavily on the level of residual kidney function.[172,173] Surprisingly, however, plasma concentrations of the bound solutes *p*-cresol sulfate and indoxyl sulfate are not much higher in patients undergoing peritoneal dialysis without residual kidney function than in those with residual kidney function, suggesting that production of these solutes may diminish as residual kidney function is lost.[173] Besides, plasma concentration of *p*-cresol sulfate and indoxyl sulfate did not fall in proportion to the increase in time-averaged clearance in the more aggressively dialyzed patient groups in either the HEMO or Frequent Hemodialysis Network (FHN) studies.[174,175] The HEMO study assessed the effect of a higher dialysis dose (target equilibrated Kt/V 1.45) and a high-flux dialyzer membrane on all-cause mortality in patients undergoing hemodialysis three times per week compared with standard dose (target equilibrated Kt/V 1.05) and a low-flux membrane. The FHN trial aimed to determine whether increasing the frequency of in-center hemodialysis would result in beneficial changes in left ventricular mass, self-reported physical health, and other intermediate outcomes among patients undergoing maintenance hemodialysis.

### SEQUESTERED SOLUTES

Some solutes are sequestered, or held in compartments where their concentration does not equilibrate rapidly with that of the plasma.[176] Application of a high dialytic clearance may rapidly lower the plasma concentration of such solutes while removing only a small portion of the total body content. When this happens, intermittent dialysis treatment will be followed by a rebound in the plasma solute concentration toward predialysis levels.[66,177] Theoretically, the contribution of sequestered solutes to uremic toxicity, like the contribution of large solutes or protein-bound solutes, could be assessed by comparing the efficacy of different dialysis prescriptions. When treatment is intermittent, the removal of sequestered relative to freely equilibrating solutes can be increased by lengthening the treatment while simultaneously reducing the plasma clearance. To date, however, while prolonged treatment has been shown to lower plasma levels of phosphate, its effect on levels of organic uremic solutes has not been assessed.[159]

## EFFECTS OF DIET AND GASTROINTESTINAL FUNCTION

It may be possible to identify uremic toxins by comparing the effect of different diets as well as by comparing the effect of different renal replacement therapies. Patients with kidney failure tend to reduce their intake of protein spontaneously.[179] Before dialysis was available, physicians found that the protein restriction could ameliorate uremic symptoms.[180] These findings suggest that important uremic toxins are derived from protein catabolism. They call into question current recommendations that patients undergoing dialysis ingest a higher protein intake than what has been recommended for the general population.[181] Uremic solutes whose production depends on protein intake include urea, MG, GSA, and the indoles and phenols produced by the action of gut bacteria on tryptophan, phenylalanine, and tyrosine.[75,86,182–184] This group overlaps with the large group of uremic solutes made by colon microbes.[134] The production of such solutes may depend not only on dietary intake but also on gut function. Impaired small bowel function may increase the delivery of peptides to the colon in uremia, and the composition of the colon microbiome may also be altered.[185,186] If colonic bacteria produce uremic toxins, uremic symptoms could theoretically be relieved by altering the delivery of substrates to the colon or adding sorbents to the diet. Only limited studies of such maneuvers have so far been performed, with success most often obtained with the use of dietary fiber to reduce the production of selected colon-derived solutes.[187–189] Historically, once hemodialysis became widely available, attempts to modify uremic solute production were largely abandoned. However, interest in this area has been revived by the imperfect efficacy of conventional dialysis and by the relatively disappointing results to date of trials evaluating more intensive dialysis treatment.[98] Interest has also been stimulated by new knowledge of the microbiome, and the promise that if toxic solutes were better known, the microbiomes composition or enzymatic function could be manipulated to reduce their production.[190]

## SOLUTE CLEARANCE BY ORGANIC TRANSPORT SYSTEMS

The cloning of transporters, which move organic solutes into the lumen of the proximal tubule, has provided a potential new means to identify potential uremic toxins. To the extent that uremia is caused by accumulation of organic solutes, knocking out these transporters would be expected to recapitulate uremic symptoms. To date, knocking out individual transporters has been found not to cause detectable illness, likely because of redundancy of the transport systems.[191,192] The accumulation of uremic solutes may interfere with organic solute transport important for detoxification at other sites, most notably the liver and blood–brain barrier.[193–195]

# METABOLIC EFFECTS OF UREMIA

The loss of kidney function has numerous metabolic effects. Some of the most prominent are listed in Box 52.1. A few can be related to the loss of specific renal processes, such as the hydroxylation of vitamin D. However, most have no clear cause, and can at present only be attributed to the retention of uremic solutes.

## OXIDANT STRESS AND THE MODIFICATION OF PROTEIN STRUCTURE

Studies have suggested that loss of kidney function increases oxidant stress.[196] The term “oxidant stress” is acknowledged to be vague, although a wealth of evidence points to increased oxidant effects in uremia. Increased levels of primary oxidants cannot be documented because they are evanescent species, which act locally, such as superoxide anion, hydrogen peroxide, hydroxyl radical, and hypochlorous acid. The accumulation of various by-products of oxidant reactions is therefore taken as evidence of increased oxidant activity. Although the accumulation of these markers of oxidant activity is well documented, there is at present no explanation as to why the production of oxidants should be increased in uremia. Leukocyte activation leading to increased production of hypochlorous acid has been described in patients undergoing dialysis and may be especially prominent when uremia is accompanied by systemic inflammation.[197]

Among the most commonly measured markers of oxidant activity have been oxidized amino acids and related compounds and substances that react with thiobarbituric acid, including malondialdehyde.[198] The accumulation of these low-molecular-weight compounds could reflect reduced renal clearance as well as increased production. More convincing evidence of oxidant stress is the accumulation of intact proteins containing oxidized amino acids.[199,200] The accumulation of these larger markers of oxidation cannot be attributed to reduced renal clearance. Further potential evidence of oxidative stress in uremia is the loss of extracellular reducing substances. The extracellular compartment is normally provided with several reducing substances, of which the reduced forms of ascorbic acid and plasma albumin are considered to be the most important. In uremia, the portion of ascorbic acid and albumin circulating in the oxidized form is increased. The case of albumin, which undergoes oxidation at its single free cysteine thiol (SH) group, is particularly interesting. Plasma albumin in patients with kidney failure is rapidly restored to the reduced form during hemodialysis.[201] The shift to oxidized albumin in untreated uremia is associated with the accumulation of cystine, which is the oxidized form of the thiol amino acid cysteine, and the shift back to reduced albumin during hemodialysis is associated with a lowering of cystine levels toward normal. One explanation for these phenomena is that normal kidney function is required to accomplish the steady reduction of cystine and albumin, which must take place to offset normal oxidant production.

The major ill effect of increased oxidant activity in uremia is thought to be modification of proteins. Proteins are modified not only by direct oxidation of amino acids but also by the combination of amino acid side chains with carbonyl (C=O) compounds. The terminology in this area is confusing. The first carbonyl compounds shown to react with proteins were sugars, and the modified proteins formed after several reaction steps were therefore referred to as advanced glycosylation end products (AGEs). Elevated sugar concentrations could account for the increased concentrations of AGEs found in persons with diabetes mellitus, but not for the subsequent findings of similarly increased AGE concentrations in patients with ESRD. Studies have shown that the high levels of active nonsugar carbonyls are responsible for the increased production of these modified proteins when renal function is reduced.[202] The active carbonyls have not been fully characterized, but they include compounds such as glyoxal (Fig. 52.1), which can be produced by oxidation of sugars and lipids. It has therefore been suggested that the protein end products of carbonyl modification in uremia should be referred to not as AGEs but as advanced glycoxidation and lipoxidation end products. Terminology aside, interest in both directly oxidized and carbonyl-modified proteins has centered on the possibility that alterations in protein structure contribute to uremia.[203,204] The hypothesis that oxidant stress contributes to adverse health consequences in ESRD has prompted trials of various antioxidants. So far, administration of vitamin C, various forms of vitamin E, folate, and α-lipoic acid has failed to reverse plasma indices of oxidant stress in patients undergoing dialysis.[205–210]

## EFFECTS OF UREMIA

### ON RESTING ENERGY EXPENDITURE

Resting energy expenditure has been reported as increased, decreased, and normal in patients with kidney failure.[211–215] The choice of control populations and other methodologic issues, such as corrections for altered body composition, have probably contributed to this uncertainty. The effect of dialysis treatment, which may transiently increase energy expenditure, must also be considered. Uremia, apart from dialysis, likely reduces resting energy expenditure.[213,215] Lower energy expenditure accords with observations of lower body temperatures in those with uremia, although additional factors may be at play in thermoregulation.[13] However, dialysis treatment may itself speed metabolism and increase energy expenditure.[214] Effects of inflammation in patients with ESRD add further complexity to the assessment of energy requirements.[216] Using anthropometric techniques in patients with ESRD on chronic hemodialysis who were maintained for 3 months on carefully constructed diets, the energy requirements were found not to differ from those reported for similarly sedentary normal subjects.[217] The energy requirements in ESRD were, however, more variable than in normals.

Knowledge of the physiologic control mechanisms for appetite, fat metabolism, and energy expenditure is potentially relevant to uremic energy metabolism. Studies have focused on the signaling molecules ghrelin, produced by the stomach, and leptin, produced by adipose tissue along with other adipose tissue–derived hormones (adipokines). Levels of these small proteins tend to rise in kidney failure because of reduced clearance by the kidney and possibly because of increased production.[54,218]

### ON CARBOHYDRATE METABOLISM

Insulin resistance is the most conspicuous derangement in uremic carbohydrate metabolism.[219] The defect is clearly present in ESRD, but in cross-sectional studies, impairment can be detected when the GFR falls below 50 mL/min/1.73 m,[2] with a graded relation to GFR.[220,221] There are probably several causes of this phenomenon.[222] Surprisingly, some obvious possibilities do not seem to contribute. Insulin binds normally to its receptor in uremia, and receptor density is unchanged.[223,224] Moreover, excess levels of glucagon or fatty acids do not account for the disorder.[225] As is the case with overall energy metabolism, interest has recently focused on the contribution of adipokines, including leptin, resistin, and adiponectin to insulin resistance. Adipose tissue has also been identified as a source of inflammatory cytokines that impair insulin action in various experimental systems and circulate at increased levels in many patients with advanced renal insufficiency.

However, correlations among levels of individual substances with measures of insulin resistance are poor, and the extent to which adipose tissue products contribute to uremic insulin resistance remains uncertain.[55,225,226]

Because dialysis, transplantation, and low-protein diets tend to restore insulin responsiveness, it has also been suggested that unidentified nitrogenous product(s) mediate insulin resistance.[225] Acidosis has been shown to provoke insulin resistance and accumulation of acid, as well as nitrogenous wastes, may contribute to insulin resistance in uremia.[227] 11-β-Hydroxysteroid dehydrogenase type 1 provokes insulin resistance by regenerating glucocorticoids. In experimental uremia, liver and fat tissues express increased activity of this enzyme as well as insulin resistance. The insulin resistance associated with 11-β-hydroxysteroid dehydrogenase can be mitigated with an inhibitor of the enzyme, all suggesting a role for this steroid pathway in insulin resistance.[228] The cause for increased enzymatic activity is unknown. Resistin is a protein capable of inducing insulin resistance, and its levels are high when kidney function is impaired. However, the plasma concentrations of resistin are not associated with insulin resistance if the GFR is taken into account.[55] Retinol-binding protein is also associated with insulin resistance (and also rises in ESRD) but is not related to markers of glucose control.[229] Finally, physical inactivity and deconditioning may contribute to insulin resistance. Exercise programs have been shown to mitigate insulin resistance but must be relatively protracted and intensive to be effective.[230,231]

Insulin resistance may have several adverse effects. Most importantly, it has been recognized as a risk factor for cardiovascular disease.[232] The connections between insulin resistance and vascular disease are not clear. A tendency to hyperglycemia is one presumably toxic effect. Some investigators have suggested that the sodium retentive effect of insulin on the kidney remains intact, whereas other tissues become insulin resistant in uremia. Increased plasma insulin concentrations could thus contribute to arterial hypertension in patients with impaired kidney function.[233,234] Outside of vascular disease, the loss of insulin's anabolic action may contribute to uremic muscle wasting.[225,235]

Even though insulin resistance is the rule in uremia, hypoglycemia can be a significant effect of renal insufficiency.[236] Hypoglycemia is likely to occur, despite insulin resistance, for two main reasons. First, the kidney is a major site of insulin catabolism. Patients with diabetes mellitus treated with insulin, or insulin secretagogues (e.g., sulfonylureas), frequently become hypoglycemic if doses are not adjusted downward as the GFR declines. Second, the kidney is a major site of gluconeogenesis.[236] The liver produces the bulk of glucose in postabsorptive and starvation states, but even in these situations, the kidney produces some glucose. With prolonged fasting, the kidney is responsible for approximately half of the total glucose production.[237,238] Thus advanced CKD may predispose to hypoglycemia, both by prolonging insulin action and by reducing gluconeogenesis. These effects may become particularly apparent when other hypoglycemic factors, such as ethanol ingestion or liver disease, are also evident.

## ON LIPID METABOLISM

Nephrotic syndrome and even lower-grade proteinuria are regularly associated with hyperlipidemia.[239] However, lipid abnormalities are modest when kidney function is impaired without significant proteinuria.[240] Indeed, total cholesterol falls on average as the GFR drops below about 30 mL/min/1.73 m$^2$.[179] Metabolite profiling in plasma of patients with ESRD using liquid chromatography and tandem mass spectrometry has revealed that lipid products deviate from normal levels far less than polar compounds. However, lower-molecular-weight triacylglycerols were generally decreased, and an increase in intermediate-weight triacylglycerols was observed.[3] The causes and consequences of these changes in lipids are uncertain. The decline in total cholesterol is taken to reflect, at least in part, progressive reduction in food intake. Numerous hypotheses have been advanced to account for the finding that atherosclerosis is accelerated but lipid levels are not markedly elevated in renal failure. Even though total lipid levels are not significantly elevated, there may be an increase in oxidized forms due to oxidant stress and reduced lipoprotein clearance rates.[241] The high prevalence of cardiovascular disease patients with ESRD has prompted several randomized clinical trials of statin agents. The largest trials performed to date have identified no clear benefit of statins in patients on dialysis, in sharp contrast to persons with normal or near-normal kidney function and, in one study,[242] to patients with advanced, non–dialysis-requiring CKD.[243–245] These trials are discussed in more detail in the chapter on cardiovascular disease (Chapter 54). Furthermore statins seem to provide no benefit in slowing the progression of kidney disease.[246]

## ON AMINO ACID AND PROTEIN METABOLISM

The normal kidney participates in the metabolism of several amino acids.[8,247–250] For example, the kidney converts citrulline to arginine. Loss of this function likely contributes to the increasing ratio of citrulline to arginine as the GFR declines below 50 mL/min/1.73 m$^2$.[248,249] Similarly, reduced renal production of serine from glycine probably underlies the rise in the plasma glycine-to-serine ratio. Increased concentrations of the sulfur-containing amino acids—cystine, taurine, and homocysteine—are especially intriguing. Cystine and homocysteine accumulate in the oxidized form, consistent with the concept that uremia is a state of oxidant stress, and homocysteine levels have been associated with the progression of cardiovascular disease.[248,249,251] Administration of folate to lower homocysteine levels neither restores the plasma redox state toward normal nor reduces the frequency of cardiovascular events.[207,252] The mechanism responsible for the accumulation of oxidized cystine and homocysteine remains unclear, but this change can be detected as the GFR drops below roughly half of normal and becomes more extreme at ESRD approaches.

Tissue protein loss reflected by muscle wasting is a major concern in patients with kidney failure. Factors that predispose to protein wasting include reduced appetite, along with insulin resistance and altered amino acid metabolism, as described earlier. Dialysis also results in some protein loss, with amino acids lost in the hemodialysate and plasma proteins and amino acids lost in the peritoneal dialysate. In the absence of other complications, the effect of uremia at the levels now seen clinically on protein metabolism is usually modest, at least over the short term.[235] Patients with advanced CKD can maintain nitrogen balance on low-protein diets as long as acidosis and inflammation are minimal. Several factors may combine with defects in insulin resistance and altered amino acid and adipose metabolism to produce muscle wasting. The best studied of these is acidosis, which has been shown to stimulate the ubiquitin–proteasome pathway of intracellular protein degradation. Activation of caspase-3 seems to be an important step in proteolysis, which is followed by disposal of protein cleavage fragments through the proteasome.[253] In addition to these effects, acidosis contributes to insulin resistance and thereby attenuates protein anabolic actions of insulin.[254] Base supplements can mitigate the catabolic effects of acidosis, but a long-term study establishing the value of normalizing bicarbonate levels in patients with impaired kidney function is lacking.[227,255–257]

Inflammation may be an even more important contributor to protein wasting than acidosis in patients with kidney failure. Muscle loss in patients with ESRD has been linked to an inflammatory state, characterized by increased serum concentrations of C-reactive protein and various cytokines. How these inflammatory mediators trigger net protein degradation in muscle and other tissues remains to be elucidated, although their presence is regularly accompanied by muscle loss. High levels of inflammatory mediators are also associated with lower serum albumin concentrations, which have been attributed largely to reduced hepatic production of albumin. Both muscle loss and reduced albumin concentration predict early death. The exact cause(s) of inflammation remains elusive. In some cases, inflammation can be ascribed to known episodes of infection or other intercurrent illness. In many cases, however, no cause can be identified. Occult inflammatory stimuli in these cases may include subclinical infection at hemodialysis catheter or arteriovenous graft sites, exposure to dialysate and various synthetic materials, and accelerated vascular injury that is common in ESRD.[258–260] Oxidant stress has also often been invoked. Attempts to reduce inflammation with free radical scavengers and other agents, however, have been largely unsuccessful.[261–263] An interesting possibility is that organic solutes retained in kidney failure, although they do not regularly trigger inflammation over the short term, cause the late appearance of inflammation in a subset of patients with ESRD. Some evidence has suggested that accumulation of proteins modified by glycation and oxidation can trigger a self-perpetuating inflammatory loop in these cases.[264,265]

Another factor contributing to muscle wasting is inactivity. Johansen and associates have shown that self-reported physical activity in patients starting dialysis is at, or below, the first percentile for population reference range,[266] and lower levels of physical activity are strongly associated with mortality.[267] In these patients, inactivity may be caused by fatigue and loss of energy, which are invariable although difficult-to-measure features of uremia, and by depression and other comorbid illnesses,[268] which are common in the ESRD population.

### OVERALL NUTRITION

As emphasized by Depner,[9] the condition of patients undergoing dialysis reflects a combination of residual uremia, side effects of dialysis mixed with the effects of comorbid conditions, and increasing age. Most patients starting on dialysis in Europe and the United States are overweight. This reflects population-wide overeating, which can contribute to, and accelerate the progression of, CKD. The protein wasting exhibited by a subset of patients undergoing dialysis is thus not malnutrition in the sense of limited nutrient availability. It is often accompanied by anorexia and reduced food intake, particularly when inflammation is prominent. It cannot, however, be reversed simply by increasing food intake.[269] Other measures of restoring body protein and muscle mass toward normal, including exercise, appetite stimulants, and newer anabolic and antiinflammatory agents, are under study.

## SIGNS AND SYMPTOMS OF UREMIA

Frequently identified signs and symptoms of uremia are listed in Box 52.1. That a fundamental metabolic disturbance such as uremia should have such a wide variety of consequences is not remarkable. The complications of untreated diabetes or hyperthyroidism are similarly extensive. However, uremia is different in that we cannot trace all its complications to dysregulation of a single key compound. And, except for renal transplantation, current therapy for uremia cannot return patients as close to normal as thyroid hormone or insulin replacement.

The level of renal function at which uremia can be said to appear is obscure. Furthermore, the diminution of functions other than solute clearance likely contributes to the symptoms and signs of uremia. In general, these other functions, such as ammoniagenesis, erythropoietin, and 1,25-dihydroxyvitamin D synthesis, urine-concentrating capacity, and tubular secretion, tend to decline in parallel with GFR, but not always. Nevertheless, defining the level of kidney function solely by GFR may be misleading. For example, certain potentially toxic solutes depend more on tubular secretion than on glomerular filtration for their excretion, and renal synthetic processes are probably linked to GFR only by virtue of the loss of functioning renal tissue. However, until particular renal dysfunctions are attached to specific aspects of the uremic syndrome, GFR will remain the principal index of kidney function.

Most of the clinical and biochemical characteristics of uremia have been defined in ESKD or at a level of GFR very near to ESKD. Thus, as noted at the beginning of this chapter, uremic characteristics may be hard to dissect from complications of the dialysis procedure. Other morbidities considered separate from the uremia also commonly interact with it. For example, the cardiovascular disease suffered especially by patients with diabetes and hypertension appears to be accelerated by CKD. However, the myocardial infarctions, strokes, and peripheral vascular diseases suffered by these patients have not traditionally been considered features of the uremic syndrome. These conditions nevertheless add to patients' disabilities in ways that are often not easily distinguishable from uremia or the residual syndrome of ESKD. Similarly, the peripheral neuropathy and gastroparesis of diabetes are difficult to disentangle from uremic neuropathy and uremic anorexia, nausea, and vomiting.

### WELL-BEING AND PHYSICAL FUNCTION

Given the list of signs and symptoms in Box 52.1, it is not surprising that health-related quality of life (HRQOL) tends to decline in patients with CKD. The point in the course of

CKD at which quality of life begins to decline has not been dissected in great detail, but some data exist. The authors of the National Kidney Foundation Kidney Disease Outcomes Quality Initiative (NKF KDOQI) guidelines have concluded that notable reductions in well-being appear when the GFR is <60 mL/min/1.73 m$^2$. Dialysis undoubtedly imposes a burden on patients. Interestingly, however, comparisons of HRQOL in patients on dialysis and patients with advanced CKD who are not on dialysis have yielded discordant results.[270–273] Other, often neglected features of treatment, such as pill burden and frailty, may also contribute to reduction in the quality of life.[274] Patients with ESKD are on average more depressed than healthy controls. However, it is difficult to distinguish the extent to which depression is caused by uremic solutes as compared with the effects of comorbid disease and the knowledge of ill health and limited life expectancy. Not surprisingly, transplantation has been found to consistently improve quality of life.[275] However, early versus later initiation of chronic hemodialysis does not seem to influence quality of life.[276]

Physical functioning in patients treated with dialysis is decidedly below normal.[277] The self-reported activity of people initiating dialysis is below the fifth percentile for healthy people.[266] Treatment of anemia improves this situation but does not normalize it.[278,279] The most detailed studies have identified multiple defects that are associated with fatigability.[280] These include muscle energetic failure and neural defects. The degree to which they are attributable to the uremic environment itself, deconditioning, and/or comorbid conditions such as diabetes mellitus is difficult to establish. Even highly functional patients on dialysis display notable physical limitations. Blake and O'Meara[281] have reported that middle-aged patients undergoing dialysis, with good nutrition and no significant comorbidities, exhibit a wide range of quantifiable deficiencies. For example, balance, walking, speed, and sensory function in these patients were clearly below those of matched controls.

## NEUROLOGIC FUNCTION

A particularly interesting group of uremic signs and systems reflects altered nerve function. Classic descriptions emphasized that uremic patients could appear alert, despite defects in memory, planning, and attention.[13,282] As kidney function worsened, patients progressed to coma or catatonia, which could be relieved by dialysis. Today, patients maintained on dialysis exhibit more subtle cognitive defects.[283] A difficulty in identifying the effects of uremia in these patients is that the hemodialysis procedure and/or associated factors (e.g., hypotension) may transiently impair cognitive function.[284] Studies in patients with CKD have suggested that cognitive impairment can be detected when the GFR falls below 60 mL/min/1.73 m$^2$ and worsens as the GFR declines.[285–287] As with other signs and symptoms of uremia, the degree to which cognition is influenced by uremia, as opposed to other comorbidities, especially cerebrovascular disease, is difficult to ascertain. Imaging studies suggest that subclinical cerebrovascular disease is common in CKD, and its role in poor cognition needs further definition.[288–290] The finding that kidney transplantation improves cognitive function suggests, however, that at least some of the impairment observed in ESRD patients is reversible and, perhaps, due to solute accumulation.[291–293] Impaired cognitive function has so far been associated with accumulation of one specific uremic solute, 4-hydroxyphenylacetate, and been shown to be partly reversed by acute dialysis treatment.[112,294] A further reflection of altered CNS function in uremia is impaired sleep.[295,296] Sleep is fragmented by brief arousals and apneic episodes, which are often associated with bursts of repetitive leg movement. When awake, patients may feel a need to move their legs continuously, termed the "restless legs syndrome."[297,298]

Sensorimotor neuropathy was a recognized component of the uremic syndrome decades ago.[13] Studies of conduction velocity and other nerve functions have since repeatedly found that most patients with uremia have peripheral neuropathy, albeit often subclinical.[282,299,300] Morphologic studies have shown that these functional changes are associated with axonal loss, and therefore may not be reversible in some patients. The extent to which peripheral nerve function is impaired earlier in the course of CKD is not certain. Autonomic neuropathy also develops in ESRD, but has been less extensively studied than peripheral neuropathy.[300] As with other uremic disturbances, the cause of neuropathy is unknown. Parathyroid hormone, multiple retention solutes, and more recently potassium have been associated with peripheral neuropathy, but without definitive proof of causality.[282,299]

## APPETITE, TASTE, AND SMELL

Loss of appetite is a common uremic symptom and presumably contributes to malnutrition in patients with advanced renal failure. A large number of causes have been proposed. Acidosis and inflammatory cytokines, including tumor necrosis factor and various interleukins, have been identified as contributing factors.[301] As with the uremic defects in energy metabolism, attention has been focused on the accumulation of small proteins produced by the gut and adipose tissue and acting on the brain to regulate appetite in normal people.[302,303] Levels of leptin, an anorexigen produced by adipose tissue, are elevated in ESKD. Antagonism of leptin in mice with experimental CKD attenuates a number of the molecular markers of proteolysis.[304] An interesting feature of uremic anorexia that remains to be explained is a disproportionate reduction in the intake of protein.[179] Along with overall loss of appetite, erosion of taste and smell has long been recognized in the ESRD population.[305,306] As with most defects, transplantation reverses the blunted smell.[305] Some studies have reported that odor threshold declines gradually with creatinine clearance, whereas others have found that even in patients undergoing dialysis, odor detection remains normal unless malnutrition is present.[143,305] Taste acuity has been reported as lower in patients undergoing dialysis than in those with renal insufficiency, and self-reported altered taste is associated with poor nutritional status.[307,308] The factors responsible for these defects are again unknown.

## CELLULAR FUNCTIONS

The most general cellular abnormality reported has been the inhibition of sodium–potassium adenosine triphosphatase ($Na^+,K^+$-ATPase). Decreased $Na^+,K^+$-ATPase activity in red cells of uremic patients was reported in 1964.[309] In general, subsequent reports have confirmed the observation, noted the same effect in other cell types, and emphasized that the inhibition was attributable to some factor in uremic serum.[310] The evidence for a circulating inhibitor includes the findings that dialysis reduces the inhibitory activity and uremic plasma can acutely suppress the pump activity.[310] However, the factor or factors have remained elusive. A number of candidates have been considered. Much attention has focused on digitalis-like substances. Several such compounds have been found in excess in humans with ESRD. These include marinobufagenin and telocinobufagin, which have a structure related to that of digitalis.

## WHY IS THE GLOMERULAR FILTRATION RATE SO LARGE?

Glomerular filtration, the initial step in urine formation, is quantitatively huge, with a volume equaling that of the entire ECF filtered every 2 hours. At rest, approximately 10% of the body's energy consumption is devoted to reabsorption of valuable solutes and water necessitated by this massive filtration rate. The rate of fluid processing clearly exceeds that required to rid the body of the daily intake of water and inorganic ions. Theoretically, the large tubular flow rate provided by the GFR could supply a sink into which organic solutes are secreted more favorably than at lower tubular flows. This hypothesis accounts for the presence of a large GFR but leaves unanswered the question of which solutes must be handled by secretion and thereby maintained at a low level in the ECF.

Homer Smith (1895–1962) recognized that the mammalian GFR was large in proportion to the kidney's known functions. He suggested that our high GFR was an evolutionary residual of the mechanism that allowed early vertebrates living in freshwater seas to excrete large volumes of water. If this were correct, the superfluity of GFR would constitute an expensive vestige in land-dwelling mammals, and the value of tubular secretion would remain unaccounted for. An alternate explanation for the apparent superfluity of kidney function is that it provides a safety factor, similar to the capacity of bone to withstand greater than usual mechanical loads. In the case of the kidney, the ingestion of toxins could constitute an analogous increase in load. It is noteworthy, however, that the proportion of GFR and kidney size to metabolic rate appears to be nearly constant across mammalian species, including herbivores and carnivores.[311,312] This suggests that the substances with excretion that necessitate a large kidney are products of common metabolic pathways rather than specific foodstuffs. The remarkable ability of bears to reduce kidney function and net protein breakdown to near zero during winter denning further suggests that these substances are end products of protein catabolism.[313,314]

We can further suppose that kidney capacity appears excessive because our clinical criteria are too coarse to detect the consequences of mild impairment in kidney function. Fitness in an evolutionary sense may require the concentrations in body water of some excreted solutes to be maintained below the levels at which we detect disease. That is, our clinical criteria for uremic illness may be too coarse to detect the consequences of mild impairment of renal function. One might speculate that disturbances in an important but sensitive parameter, perhaps fertility, growth in children, or peak physical performance, would occur with less than a twofold increase of some retained toxin. A particularly interesting finding has been the identification of similar transport systems in the kidney tubule and blood–brain barrier.[194,315] This finding suggests that the kidney, together with the liver, may be designed to keep organic waste levels in the ECF sufficiently low so that a second-stage pumping system in the blood–brain barrier can keep the brain interstitium exquisitely clean.

Complete reference list available at ExpertConsult.com.

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